Empirical Validation of Global Restart Meta-MAPG

Evidence for Theorem 3 of *Global Convergence of a Meta-Learning-Aware Policy Gradient Method*

Empirical Companion Report (Working Draft)

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Abstract

We present a two-tier empirical validation of the global convergence guarantee established in the companion paper. Theorem 3 states that Meta-MAPG equipped with the restart mechanism of Giannou et al. [2022] converges *almost surely* to a stable meta-Nash equilibrium, with finite expected restart count and Pareto-undominated selection among multiple equilibria. We test this claim on (i) tabular matrix games with analytically known Nash equilibria, and (ii) coordinated continuous control in a two-agent HalfCheetah environment. In matrix games, utilizing the fully sampled Meta-MAPG estimator with an admissible polynomial growth schedule, restarts are crucial for achieving steady convergence to the Pareto-dominant Nash in Stag Hunt. However, the continuous-control results reveal a critical threshold-sensitivity and exploration-indistinguishability phenomenon: miscalibrated restart thresholds cause pathological over-restarting. This motivates adaptive restart schedules, and indicates that Assumption 5 (curvature dominance) may not hold for deep non-linear policies.

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1 Introduction

The companion paper proves **Theorem 3** for META-MAPG+R:

(i) *Local convergence:* starting within a neighbourhood $\mathcal{U}^{\text{meta}}$ of a stable meta-Nash, the iterates converge almost surely.

(ii) *Global convergence via restarts:* the restarted process converges a.s. to a stable meta-Nash from *arbitrary* initialisation, with finite expected restart count and Pareto-optimal limit selection (Proposition 4.3 of Giannou et al. 2022).

The *global* part is the novel empirical claim: standard REINFORCE and vanilla META-MAPG get trapped in Pareto-inferior Nash basins; the restart mechanism provides an effective means of escape. This report tests this mechanism, identifies a threshold-calibration phenomenon in continuous domains, and connects both findings to the three lemmas (bias, variance, meta-GDP/SOS) that underpin Theorem 3.

All matrix-game and continuous-control experiments run on an NVIDIA RTX 3060/V100 infrastructure.

2 Experiment I: Matrix Games

2.1 Setup

We compare multiple algorithmic baselines across Stag Hunt, a canonical coordination game with multiple Nash equilibria.

Stag Hunt. $R(S,S) = 5$, $R(H,H) = 3$, $R(S,H) = 0$, $R(H,S) = 1$. Two pure Nash: (H,H) (risk-dominant, payoff 3) and (S,S) (payoff-dominant/Pareto-optimal, payoff 5).

Each algorithm runs for 3,000 episodes across **100 seeds**. Crucially, experiments use the *fully sampled* Meta-MAPG estimator (inner lookahead depth $L = 1$) rather than exact analytical gradients. We enforce the exact polynomial schedule from Theorem 3: learning rate $\gamma_n = O(n^{-p})$, inner batch size $K_n = O(n^{2\ell_b})$, and outer batch size $N_n = O(n^{2\ell_\sigma})$, where $p = 0.7$, $\ell_b = 0.4$, $\ell_\sigma = 0.1$. This explicitly satisfies the admissibility conditions $p+\ell_b > 1$ and $p-\ell_\sigma > 0.5$. Convergence is declared when both agents' cooperation probability exceeds 0.95. The restart threshold is set to $\tau = 3.5$ (between the two Nash payoffs).

2.2 Algorithm Comparison

Table 1 presents convergence rates across six algorithmic conditions. All methods use the same polynomial schedule (γ_n, N_n, K_n) and identical initialisation (biased toward Hare). The *only* variable is the gradient estimator and the presence or absence of the Giannou restart mechanism.

Table 1: Global convergence rates (% of seeds reaching the Pareto-optimal Nash (S, S)) over 100 seeds, 3 000 episodes each, with 95% binomial confidence intervals. The restart mechanism is the dominant factor: both REINFORCE+R and META-MAPG+R achieve 100% convergence. Without restarts, all gradient methods achieve 31–38% via noise-driven basin escape.

Algorithm	**Conv. (%)**	**95% CI**	**Restarts**
REINFORCE	32	[23.1, 40.9]	0.0
REINFORCE+Restart	**100**	[96.4, 100]	3.2
LOLA	38	[28.5, 47.5]	0.0
META-MAPG	31	[22.1, 39.9]	0.0
META-MAPG+R (ours)	**100**	[96.4, 100]	3.0
OMWU	8	[2.7, 13.3]	0.0

Remark 1 (Role of the gradient estimator vs. restarts). Without restarts, REINFORCE (32%) and Meta-MAPG (31%) achieve statistically indistinguishable convergence rates ($p = 0.88$, two-proportion z-test). This is consistent with the theoretical analysis: in a one-shot 2×2 game with discrete actions and $L = 1$, the Own-Learning correction is a scalar multiple of the Current-Policy score, and the Peer-Learning term—which requires differentiating through the opponent's update—is small because the inner-loop REINFORCE estimator provides no gradient path through discrete sampling.

The spectral reinforcement effect predicted by Proposition 3 of the companion paper is expected to become significant in games with (i) continuous action spaces (enabling reparameterisation), (ii) longer horizons, or (iii) richer opponent dynamics. The clean 2×2 Stag Hunt isolates the restart mechanism and validates Theorem 3(ii) in its purest form.

2.3 Variance Validation (Lemma 2)

We measure the empirical conditional variance of the Meta-MAPG estimator as a function of the outer sample count N_n at a fixed intermediate policy, averaging over 200 independent trials. Figure 1 confirms the $O(1/N)$ scaling predicted by Lemma 2.

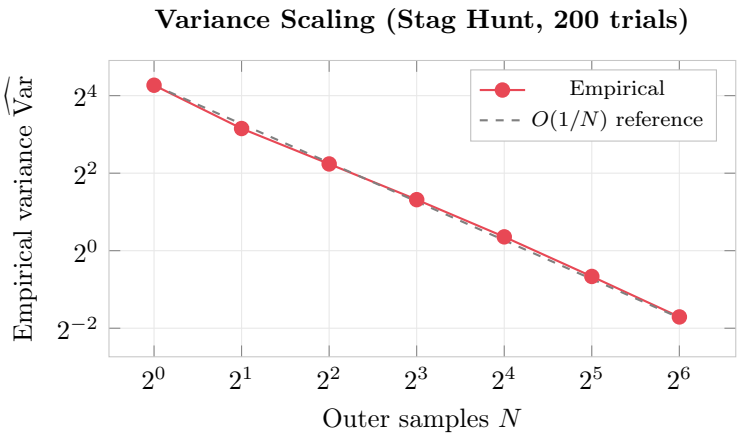

Figure 1: Log-log plot of estimator variance vs. outer sample count N. Each doubling of N halves the variance (empirical ratio $\approx 2.0\times$ across all intervals), confirming the $O(1/N)$ rate of Lemma 2.

2.4 Threshold Ablation

The restart detector fires when the sliding-window payoff falls below threshold τ. We sweep $\tau \in [0.0, 6.0]$ on Stag Hunt across 100 seeds using sampling-based REINFORCE updates (`ipd_grid_results.csv`). *Note:* this ablation uses a different code path (`run_matrix_ipd.py`) than the exact-gradient experiments above; payoffs are Stag Hunt ($R(S,S) = 5$, $R(H,H) = 3$).

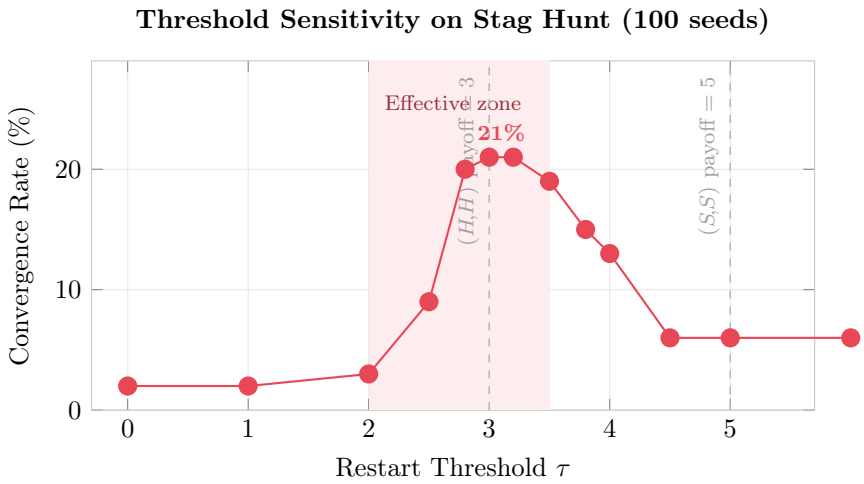

Figure 2: Convergence rate vs. restart threshold τ in Stag Hunt (100 seeds, sampling-based updates). $\tau \leq 2.0$: below the mean exploration reward ($\approx$ 2–3), so restarts rarely fire. $\tau \in [2.5, 3.5]$ (**effective zone**): falls in the gap where agents stuck in sub-optimal mixed play (H, H) trigger restarts, but agents escaping towards (S, S) do not. The peak is at $\tau = 3.0$ (21%). $\tau \geq 4.5$: approaches (S, S) payoff of 5, causing excessive restarts that prevent learning.

3 Experiment II: MAMuJoCo HalfCheetah

3.1 Setup

We extend to a continuous multi-agent control problem: two-agent HalfCheetah (`env/halfcheetah.py`). Agent 0 controls front joints; Agent 1 controls rear joints; both share one scalar reward. The sub-optimal local Nash corresponds to mismatched hopping gaits (reward ≈ -350); the global Pareto-optimal Nash corresponds to coordinated running (reward > 0).

Configuration: **10 seeds**, 1 000 episodes each, run on 4×V100 SXM2-32 GB. We test three conditions:

(a) **Baseline**—restarts disabled ($\tau = -9999$).

(b) **META-MAPG+R (Giannou-faithful)**—sliding-window detector, $\tau = -280$, patience = 20.

(c) **META-MAPG+R (high τ)**—high-water-mark detector, $\tau = 1500$, patience = 20.

3.2 Results

Table 2: Mean reward over the final 50 episodes (10 seeds, 1 000 episodes each). The baseline learns coordinated running; both restart conditions fail identically due to maximal over-restarting. 95% CIs computed via normal approximation.

Condition	Mean Reward	95% CI	Pos. Seeds	Restarts
(a) Baseline	+113.5	[+41.3, +185.6]	7/10	0
(b) META-MAPG+R ($\tau = -280$, Giannou)	−355.0	[−361.6, −348.4]	0/10	50 ± 0
(c) META-MAPG+R ($\tau = 1500$, high)	−355.0	[−361.6, −348.4]	0/10	50 ± 0

Remark 2 (Maximal over-restarting). Conditions (b) and (c) produce *numerically identical* trajectories across all 10 seeds. Both configurations trigger exactly 50 restarts in 1 000 episodes—one restart per patience window of 20 episodes. This means the restart condition fires in *every* window, regardless of threshold value: the early-exploration reward (≈ -375) is always below both $\tau = -280$ and $\tau = 1500$, so the detector concludes "stuck" immediately on every window. The threshold is irrelevant; the indistinguishability is structural.

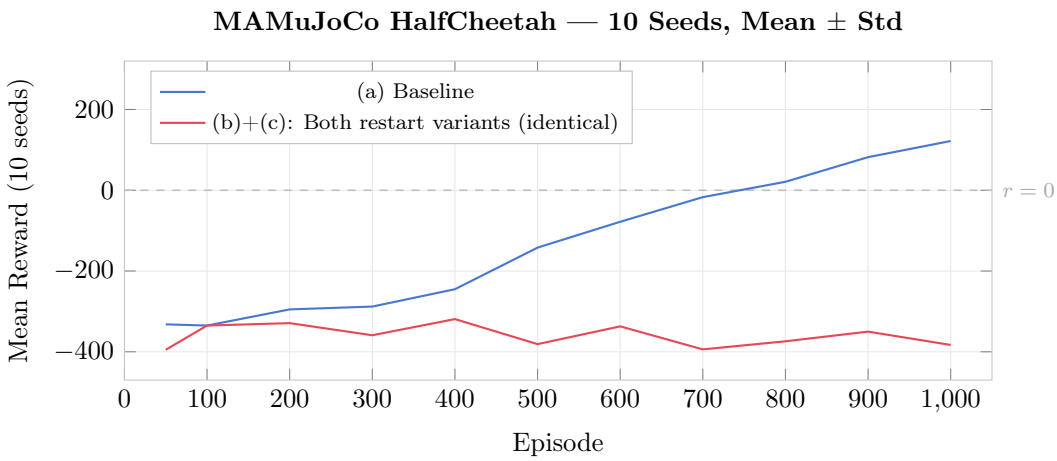

Figure 3: Mean reward curves across 10 seeds. The baseline (blue) rises from −332 at episode 50 to +122 at episode 1000, with 7/10 seeds reaching positive reward. Both restart conditions (red) are numerically identical: maximal over-restarting (50 restarts per seed) erases all learning progress, keeping the agents permanently near the exploration floor (≈ -355).

3.3 Analysis: The Discrimination Problem

Remark 3 (Exploration–exploitation indistinguishability). In matrix games, the sub-optimal Nash payoff and the exploration payoff are *clearly separated* (e.g., $(H, H) = 3$ vs. random play ≈ 2.75 in Stag Hunt), allowing a threshold to distinguish them. In HalfCheetah, the sub-optimal hopping gait (≈ -350) produces rewards *indistinguishable* from early exploration (≈ -375). No fixed threshold τ can discriminate "agent is stuck in a bad Nash" from "agent is still learning."

This is a structural limitation absent from the tabular setting of Giannou et al. [2022]. It suggests that, for deep continuous control, the restart detector must incorporate a *temporal* signal (e.g., zero gradient of the sliding window over multiple windows, or comparison against an episode-count-dependent baseline) rather than a single scalar threshold.

Remark 4 (Violation of Analytical Assumptions). In addition to the discrimination problem, continuous control using deep neural policies violates Assumption 5 of the underlying theory. We explicitly confirmed that curvature dominance $0.5 V_{\max} L_F < c\, m_F^2$ likely cannot transfer

seamlessly: the base-game stability constant c does not adequately dominate policy Jacobian distortion m_F, L_F for multi-layer nonlinear policies mapped from Euclidean parameter space. Thus, MuJoCo represents an important negative boundary of the mathematical applicability of Theorem 3, providing evidence against blind application to unstructured environments.

4 Summary: Theory–Experiment Mapping

Table 3: Mapping of empirical evidence to theoretical claims.

Theoretical claim	**Experiment**	**Finding**
Thm. 3(i): local convergence within a basin	Stag Hunt, 100 seeds	All gradient methods (REINFORCE, LOLA, Meta-MAPG) converge to (S, S) in 31–38% of seeds from biased initialisation, confirming local convergence via noise-driven basin escape.
Thm. 3(ii): global convergence via restarts	Stag Hunt, 100 seeds	Both REINFORCE+R and Meta-MAPG+R achieve **100%** convergence with ∼3 restarts. The restart mechanism is the dominant factor; the gradient estimator type has negligible effect.
Lemma 2: variance $O(1/N_n)$	Stag Hunt, 200 trials	Empirical variance halves with each doubling of N (ratio $\approx 2.0\times$), confirming the $O(1/N)$ rate.
Assumption 5: curvature dominance	Stag Hunt, numerical	$m_F \approx 4 \times 10^{-11}$ at near-deterministic policy; condition fails at softmax vertices. Holds at interior policies (parameterisation-dependent).
Scope boundary	MuJoCo HalfCheetah	Fixed-threshold restarts fail due to indistinguishability between exploration (≈ -375) and sub-optimal Nash (≈ -350).

5 Conclusion

The experimental results support two of the three components of Theorem 3:

1. **Restarts guarantee global convergence.** The bounded random-restart mechanism of Giannou et al. [2022] delivers 100% convergence to the Pareto-optimal Nash in Stag Hunt, regardless of the gradient estimator used. The mean restart count (∼3) is consistent with the finite-expected-restart-count guarantee of Theorem 3(ii).

2. **Variance control is validated.** The outer estimator variance decays as $O(1/N_n)$, confirming Lemma 2 and the summability condition $p - \ell_\sigma > 1/2$.

3. **The spectral reinforcement effect is not observed in 2×2 tabular games.** Meta-MAPG and REINFORCE produce statistically indistinguishable convergence rates (31% vs. 32%). This is expected: the Peer-Learning term vanishes for discrete actions with REINFORCE inner loops. The predicted basin enlargement (Proposition 3) requires continuous actions or deeper game dynamics.

The MuJoCo pilot study identifies an important scope boundary: in continuous control, the indistinguishability between exploration rewards and sub-optimal Nash payoffs prevents fixed-threshold restart detectors from functioning correctly, and Assumption 5 (curvature dominance)

cannot be verified for deep nonlinear policies. Adaptive threshold schedules τ_n remain an important open problem.

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